

Lab Assignment 8: Requirement Analysis & Design

Parts A, B, and C

Part A: Requirement Analysis (State Identification)

State	Meaning	Entry Event(s)	Exit Event(s)	Terminal?
PENDING_PAYMENT	Booking created, waiting for customer to pay.	BOOK_SERVICE	PAY, PAYMENT_FAILED, CANCEL	No
PAYMENT_FAILED_STATE	Payment attempt failed. Waiting for retry or cancellation.	PAYMENT_FAILED	PAY, CANCEL	No
PAID_PENDING_APPROVAL	Payment successful, waiting for service center acceptance.	PAY (Success)	ACCEPT_BOOKING, REJECT_BOOKING, CANCEL	No
BOOKING_CONFIRMED	Service center accepted the booking.	ACCEPT_BOOKING	ASSIGN_TECHNICIAN, CANCEL	No
TECHNICIAN_ASSIGNED	A specific technician has been assigned to the job.	ASSIGN_TECHNICIAN	DISPATCH_TECHNICIAN, CANCEL	No
TECHNICIAN_EN_ROUTE	Technician is traveling to the customer's address.	DISPATCH_TECHNICIAN	START_SERVICE, CANCEL	No
SERVICE_IN_PROGRESS	Technician has arrived and is performing the repair.	START_SERVICE	COMPLETE_SERVICE	No
SERVICE_COMPLETED	The repair is finished successfully.	COMPLETE_SERVICE	None	Yes
BOOKING_REJECTED	Service center rejected the booking.	REJECT_BOOKING	None	Yes
CANCELLED	Customer manually cancelled the booking.	CANCEL	None	Yes

Part B: State Transitions (Transition Table)

Current State	Event	Guard (Condition)	Resulting State	Action
Initial	BOOK_SERVICE	Valid details provided	PENDING_PAYMENT	Save booking data, prompt for payment
PENDING_PAYMENT	PAY	Payment logic returns Success	PAID_PENDING_APPROVAL	Log payment, notify service center
PENDING_PAYMENT	PAYMENT_FAILED	Payment logic returns Failure	PAYMENT_FAILED_STATE	Notify customer to retry
PAYMENT_FAILED_STATE	PAY	Payment logic returns Success	PAID_PENDING_APPROVAL	Log payment, notify service center
PAID_PENDING_APPROVAL	ACCEPT_BOOKING	None	BOOKING_CONFIRMED	Notify customer of acceptance
PAID_PENDING_APPROVAL	REJECT_BOOKING	None	BOOKING_REJECTED	Process 100% full refund
BOOKING_CONFIRMED	ASSIGN_TECHNICIAN	None	TECHNICIAN_ASSIGNED	Save technician details

TECHNICIAN_ASSIGNED	DISPATCH_TECHNICIAN	None	TECHNICIAN_EN_ROUTE	Notify customer of ETA
TECHNICIAN_EN_ROUTE	START_SERVICE	None	SERVICE_IN_PROGRESS	Log repair start time
SERVICE_IN_PROGRESS	COMPLETE_SERVICE	None	SERVICE_COMPLETED	Log completion, generate receipt
Various (See Part C)	CANCEL	Cancellation allowed in current state	CANCELLED	Process applicable refund
SERVICE_IN_PROGRESS / COMPLETED	CANCEL	Invalid state for cancellation	No Change (Rejected)	Show Error: "Cannot cancel ongoing/completed service"

Part C: Cancellation and Payment Policy Design

Allowed Cancellation States & Refund Policy

The system allows customers to cancel their booking at specific stages. The refund processed depends on how far the booking has progressed:

- **PENDING_PAYMENT / PAYMENT_FAILED_STATE:** Customer can cancel. *Action:* No refund necessary since no money was paid.
- **PAID_PENDING_APPROVAL / BOOKING_CONFIRMED:** Customer can cancel. The service center has not yet allocated a technician. *Action:* 100% Full Refund issued.
- **TECHNICIAN_ASSIGNED / TECHNICIAN_EN_ROUTE:** Customer can cancel. However, since the technician's time was reserved or they are already traveling, a penalty applies. *Action:* 50% Partial Refund issued.

Disallowed Cancellation States

- **SERVICE_IN_PROGRESS / SERVICE_COMPLETED:** Once the technician arrives and starts the service, the customer **cannot** cancel. Any `CANCEL` event triggered here will be rejected by the system guard.

Service Center Rejection

- If the booking is in **PAID_PENDING_APPROVAL** and the service center triggers **REJECT_BOOKING** (e.g., due to lack of parts or out of service area), the booking moves to **BOOKING_REJECTED**. *Action:* 100% Full Refund issued automatically.